FIBER OPTIC COMMUNICATION
SYSTEM PRACTICUM REPORT
UNIT 2
FIBER OPTIC SPLICING

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APPLIED BACHELOR OF INTERNET ENGINEERING TECHNOLOGY STUDY PROGRAM
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A. Purpose

1. Understand how to properly cut and peel fiber optic cables.
2. Understand how to perform fine cutting (cleaving) on optical fibers.
3. Understand how to install connectors on fiber optic ends.
4. Understand the method of fusion splicing fiber optics.
5. Understand the use of tools and instruments used for the *splicing* process.

B. Tools and Materials

1. Digital Microscope
2. Kabel Fiber Optic
3. Stripper, Cleaver
4. Connector (SC, LC, FC)
5. VSL (Virtual Fault Locator)
6. Fusion Splicer

C. Theory

Preparation of optical fibers is an important stage before the splicing process is carried out, at this stage *stripper* and *cleaver tools are used*. *Strippers* are high-precision tools that function to strip the outer protective layer of optical fibers (such as 250μm coating) accurately and smoothly without damaging or scratching the glass core inside. After the shield is peeled off, the end of the optical fiber must be cut using *a cleaver*, which is a special bladed mechanical cutting tool that applies small pressure to the fibers until the glass splits completely. This cutting process ensures that the cross-section of the fiber tip face is completely flat, has a 90-degree perpendicular angle, and resembles a mirror, which is a requirement for light transmission to propagate optimally without obstacles when connected.

To link the ends of a fiber optic cable, technicians can utilize the connector interface or *the mechanical sleeve* method. Fiber optic connectors are designed as precision connectors that are paired into adapter slots, ensuring that the two ends of the cable face each other in parallel so that there is no significant signal loss at the meeting point. Meanwhile, *mechanical sleeve* or *mechanical splice* is a mechanical splicing method to fuse the two ends of fiber optics together without the melting process. This method locks and aligns the ends of the glass pieces in a special sleeve, the inside of which is generally

equipped with *an index matching gel* that functions to transmit light propagation and suppress the optical attenuation value at the meeting gap.

As a permanent connection standard with the best damping quality, the fusion method is used through a *fusion splicer* machine. This automatic machine works by aligning the positions of the optical fiber ends under the supervision of a built-in microscope, then melting them into a whole glass unit through a high-temperature heat treatment generated from an electric arc. Once the optical fibers are successfully fused, the bare connection points become so fragile that they require additional reinforcements. The melting point is then shifted into the thermal chamber of the machine to heat the *protective sleeve*, where the heat heat-shrinks the sleeve to enclose and protect the joint area from the risk of bending or breakage.

D. Working Steps

1. Cut several types of fiber optic cables (1 m long), peel off the end of the cable (5 cm) to the fiber optic part. Clean the fiber optics using an alcoholic wipe.
2. Cut the fiber optic tip using a fiber cassette tool, to get a flat FO tip. Observe the quality of the cut using a digital microscope.
3. If the cut is not good, repeat the cleaving process.
4. Check the length size of the fiber optic, coating and jacket, adjusted to the size of the existing fast-connector. Install the fiber optic tip on the SC and LC type fast connectors.
5. Take a few pieces of fiber optic cable, peel off the end to the fiber optic. Cut the fiber optic well, clean it using an alcoholic wipe.
6. Set up the Fusion Splicer JoinWit equipment, learn the function of the buttons on the machine.
7. Splice the two ends of the fiber optic using fusion splicer equipment until you get a good level of damping.
8. Observe and write down the how/procedure of using fusion splicer equipment for fiber optic connection.
9. Repeat the fiber optic connection equipped with a sleeve protector.
10. Position/slide the sleeve protector to the center of the connection, moving the connection point to the heating chamber.
11. Once locked, press the heater button. Wait for a few moments while the plastic protector heating process takes place.
12. Observe the results of the fiber optic connection that has been protected.

E. Results & Analysis

1. Observation of Fiber Optic Cable Cutting Results

- FO 1 Cable: Yellow, Single Mode 125 µm (Cleaver)



The cutting of the yellow single-mode *fiber optic cable* is carried out by stripping the outer layer using a *stripper* until it leaves *125 µm of pure glass cladding. The final cut is then executed using* a cleaver *to cut the fiber precisely at a 90-degree angle. As seen in the digital microscope image above, this cut results in a very neat, flat, and smooth cross-section of the cut that resembles a mirror. The quality of this splinter-free perpendicular cut is a requirement for the glass core to blend perfectly with minimal damping during the splicing process.*

- FO 2 Cable: Single Drop Cable, Black, Single Mode 125 µm (Cleaver)



The cutting of a single drop cable *black* single-mode *fiber optic cable* begins with stripping a thick outer jacket using *a drop cable stripper* until it leaves a *125 µm of pure glass cladding. The final cutting is then carried out using* a cleaver to cut the fiber precisely at a 90-degree angle. As seen in the digital microscope image above, this cutting process results in a very neat, flat, and

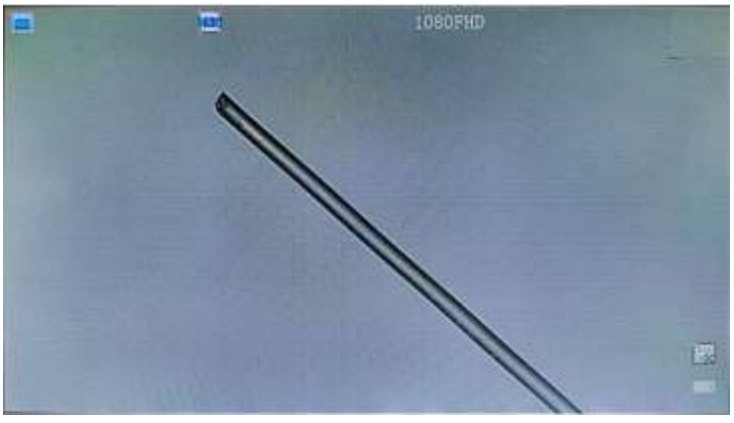
smooth cutting end cross-section resembling a mirror. The quality of this splinter-free perpendicular cut is a requirement for the glass strands to blend perfectly with the minimum level of damping during the joining process.

- FO 3 Cable: Yellow, Single Mode 125 μm (Scissors)



Cutting yellow single-mode fiber optic cables is done by stripping the outer layer using a *stripper* until it leaves *125 μm of pure glass* cladding, then fiber cutting is carried out using non-standard tools, such as scissors, which results in a very poor end cross-section. Cutting using scissors forcibly crushes and breaks the glass strands as seen in the image above. This deformed and debris cross-section is sure to refract light propagation, trigger a very high damping value, and make it impossible to put together perfectly during the splicing process.

- FO 4 Cable: Single Drop Cable, Black, Single Mode 125 μm (Scissors)

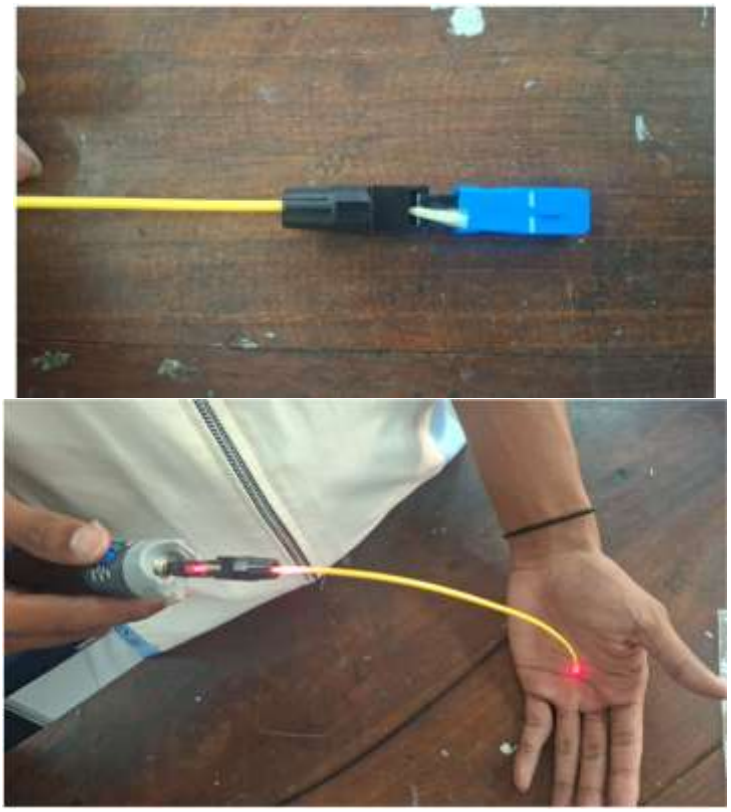


The cutting of the black *single-mode drop cable* fiber optic cable is done by stripping the outer layer using *a drop cable stripper* until it leaves *a 125 μm pure glass cladding, then cutting the fiber using non-standard tools, such as scissors, which results in a very poor end cross-section. Cutting using scissors*

forcibly crushes and breaks the glass strands as seen in the image above. This deformed and debris cross-section is sure to refract the light propagation, trigger a very high damping value, and make it impossible to put it together perfectly during the splicing process.

2. Observations on How to Install Fast FO Connectors

- Connector 1: SC Blue



The image above is the result of the process of installing a blue SC connector on a 125 μm yellow *single-mode* fiber optic cable . To check if this installation was successful, a test was carried out using *the Visual Fault Locator* (VFL) tool. When the VFL tool is plugged into the SC connector and then turned on, this tool will shoot a red laser beam that flows straight into the cable core. The appearance of bright red light spots that penetrate to the end of the cable on the other side is clear evidence that the glass inside the connector is not broken, no light is leaking, and the connector has been installed correctly.

- Connector 2: SC Green

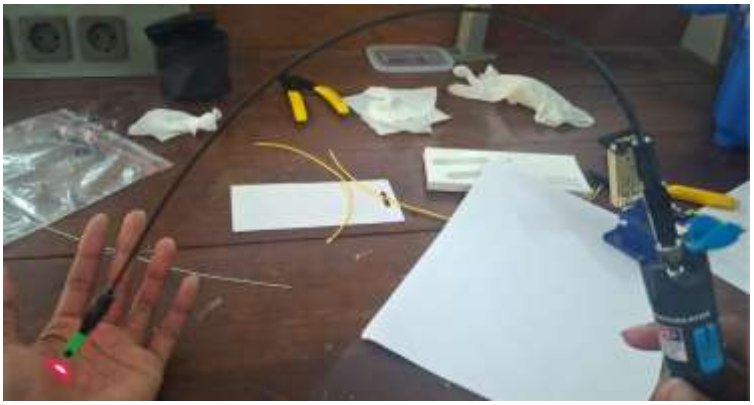




In the image above, it is the process of installing a green SC connector on a black *single-mode drop cable* fiber optic cable that requires precision so that the glass strands can enter and fit snugly in it. Unlike the blue one, this green connector has a slightly angled polished glass end, with a special function to suppress the reflection of backlight so that the signal quality remains stable and maximum. To confirm whether this installation is successful, a test was carried out using *a Visual Fault Locator* (VFL) tool. When the VFL tool is plugged into the connector and then turned on, the red laser beam will flow straight into the cable core. The appearance of bright red light spots that penetrate to the end of the cable on the other side is clear evidence that the glass inside the connector is not broken, no light is leaking, and the connector has been successfully installed.

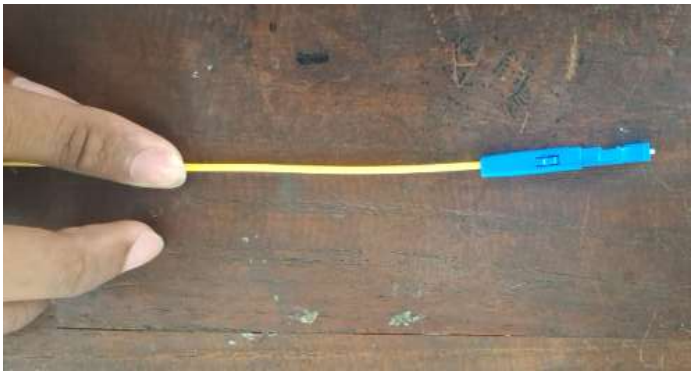
- Connector 3: FC

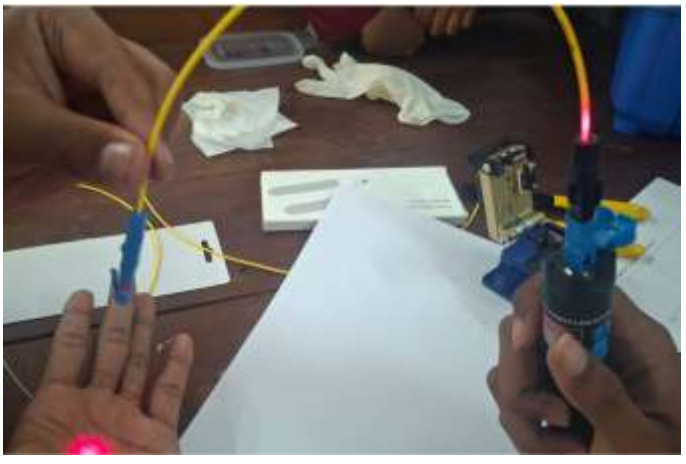




The image above shows the process of installing the FC connector on *a black single drop cable*. This connector has a metal thread locking system that functions to provide an extra strong grip, so that the connection is not easily detached by pull or vibration. The success of this installation was tested using *the Visual Fault Locator* (VFL) tool. In the image, it can be seen that the red laser beam from the VFL penetrates brightly to the other end of the cable, which is proof that the glass inside the connector is not broken, no light is leaking, and the connector is installed correctly.

- Connector 4: LC





The image above shows the process of installing the LC connector on a yellow single mode 125 um cable. This connector has a Small Form Factor (SFF) size with a locking system in the form of a clip (*retention tab*) similar to the RJ45 connector, which functions to provide a safe, stable, and easy connection to remove without requiring a large space. The success of this installation was tested using the Visual Fault Locator (VFL) tool. In the image, you can see the red laser beam from the VFL penetrating brightly to the other end of the cable, which is proof that the glass inside the connector is not broken, no light is leaking, and the connector is installed correctly.

3. Observation of Fiber Optic Fusion Splicer

- Fusion Splicer



Fusion splicer is an automation electronic device that is used to connect two ends of optical fibers permanently. Characteristically, this device is equipped with a microscopic camera system, a precision *alignment* motor, a pair of electrodes that produce electric arcs, and a heating module. Functionally, this tool is tasked with aligning and fusing the two cores into one whole path using high heat from the electric arc, so that the signal attenuation rate (*splice loss*) at the connection point is very minimal. After connection, this machine also functions to heat the *protective sleeve* to protect and strengthen the melting point so that it is not easily broken when installed.

- Connection 1



The image above is the process of permanently connecting fiber optic cables using a *fusion splicer machine*. This fusion method requires a very high level of precision so that the two ends of the glass strands can be perfectly aligned before finally being joined together using heat from the electric arc. The success and quality of this connection can be seen directly through the machine's built-in monitor screen, which can be seen in the image above. On the screen, you can see the magnification visualization of the cable core that has been tightly fused along with a very small estimated damping value (*loss*), which is clear evidence that the connection is successfully carried out optimally without obstructing the light transmission path.

- Connection 2 (Protector Sleeve Equipped)





The image above shows the process of protecting the connection points of fiber optic cables that have been fused using *a fusion splicer* machine. The fragile glass connection points are protected by sliding the *protection sleeve* to cover the connection area, then placed in the heating chamber at the top of the machine. This heating process makes the sleeve material shrink tightly clamping the cable, resulting in the final shape of a permanent connection that is wrapped stiffly, strong, and safe from the risk of fracture or bending, so that the quality of light signal transmission remains stable.

- Splicing 3 (Mechanical Sleeve)



The image above shows the results of connecting fiber optic cables using the *mechanical sleeve* method. Unlike fusion *splicer machines* that melt glass using heat, this method unites the two ends of the fiber strands manually without a melting process. The two ends of the cable that have been cut parallel are inserted into the sleeve from opposite directions until they are in close contact with each other. On the inside of the sleeve, there is generally a special gel that functions to transmit light beams while keeping the signal attenuation value low. After the position of the two ends of the glass is felt to fit, the sleeve cover is mechanically locked to clamp the cable firmly, so that the position does not shift and the connection remains stable.

F. Conclusion

- Fiber optic cables that structurally consist of coating, cladding, and core must be prepared using a stripper. This tool functions to accurately peel off the flexible jacket and outer protective layer (coating) without scratching or damaging the glass core structure inside.
- The cutting stage of the end of the glass sheet (cladding and core) uses cleaver, instead of ordinary mechanical scissors. The cleaver ensures that the cross-section of the fiber end is split perpendicular to 90 degrees, flat, and resembles a mirror, which is a requirement to avoid light refraction and high damping values.
- Connector interfaces (such as SC with blue and green color codes, LC, and FC with metal threads) are designed with a specific locking system to keep the ferrule ends precisely attached. The successful installation of these connectors is practically directly validated using a red laser beam shot from the Visual Fault Locator (VFL) to ensure no light leakage.
- Fusion splicing is a permanent splicing technique in which the ends of the optical fibers are aligned and fused into a unit using the electric arc heat of the fusion splicer machine. The fragile bare glass connection points are then strengthened and protected using a protection sleeve that is shrinked through a heating chamber.
- The entire optical infrastructure work relies heavily on two categories of equipment: special mechanical tools (strippers, cleavers, fusion splicers, or mechanical sleeves) for preparation to connection, and electronic test instruments (such as VFL) to validate the feasibility of network transmission.

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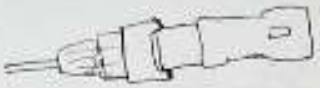
APPENDIX

C. LEMBAR PENGAMATAN :

1. Pengamatan hasil pemotongan kabel Fiber Optik

Kabel FO 1 : Kuning, single mode 125 um (versi cleaver)	Gambar hasil potongan:  Tipe cleaver :
Kabel FO 2 : Drop cable tunggal, hitam, single mode 125 um (versi cleaver)	Gambar hasil potongan: 
Kabel FO 3 : Kuning, single mode 125 um (versi gunting)	Gambar hasil potongan: 
Kabel FO 4 : Drop cable tunggal, hitam, single mode (125 um) (versi gunting)	Gambar hasil potongan: 

2. Pengamatan cara memasang fast konektor FO

FO 1 : SC Biru	Gambar : 
FO 2 : SC Hijau	Gambar : 
FO 3 : FC	Gambar : 
FO 4 : LC	Gambar : 

Cara / Alur pemasangan fast konektor:

1. Striping FO dan drop kabel tunggal
2. masukkan ujung kabel ke konektor yang sesuai
3. cek jalur kabel dgn dengan VFL
4. kunci kabel agar tidak dapat hasil yang lagas

3. Pengamatan cara penyambungan (fusion splicer) fiber optik

Peralatan/mesin fusion Splicer:	Gambar mesin/ peralatan:
Spesifikasi : • [illegible] Core Alignment dengan teknologi PAS (Profile Alignment System) • Layar LCD 5 inch • baterai Li [illegible] • Dimensi 135 mm x [illegible] mm [illegible] mm • Weight [illegible] kg	
Penyambungan #1	Gambar proses penyambungan di tampilan layar mesin:  Hasil : Loss : 0.01 dB
Penyambungan #2	Gambar proses penyambungan di tampilan layar mesin: Hasil :

Penyambungan #3 (dilengkapi protektor sleeve)	Gambar proses penyambungan di tampilan layar mesin:  Hasil : 
Penyambungan #4 (dilengkapi protektor sleeve)	Gambar proses penyambungan di tampilan layar mesin:  Hasil :

Alur /prosedur penyambungan :

1. Strip Fiber optik sekitar 2 cm (core)
2. Potong core dengan cleaver ~~[illegible]~~ dan dibersihkan dengan tisu beralkohol, supaya kabel lebih rapi dan bersih.
3. Letakkan FO ke Splicer dengan benar, lalu tutup Penutup Splicer
4. tunggu hingga kabel FO terfusion, kalau gagal error.

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